

React Master Notes - Complete Guide (50+ Pages)

Phase 1: Setup & Fundamentals

Introduction to React: Explanation, Virtual DOM concept, Why React is popular.

Setting up Environment: Node.js, npm, create-react-app, Vite.

JSX & Rendering: Syntax, Babel compilation, Differences from HTML.

Components & Props: Functional vs Class Components, Props passing, Reusability.

State Management Basics: useState hook, immutability, updating state.

Event Handling: Synthetic events, passing handlers, controlled components.

Mini Project: Build a Greeting App with dynamic messages.

Phase 2: Hooks & Lifecycle

React Hooks: useState, useEffect, useRef, useMemo, useCallback.

Lifecycle: Mount, Update, Unmount, Cleanup functions.

Custom Hooks: Creating reusable logic.

Mini Project: Build a Timer App using useEffect and cleanup.

Phase 3: Routing & Context API

React Router: BrowserRouter, Route, Link, useNavigate, Nested Routes.

Context API: Avoiding prop drilling, creating global state.

Mini Project: Multi-page Blog App with Context for theme switching.

Phase 4: Redux Toolkit & Performance Optimization

Redux Toolkit: Store, Slice, Actions, Reducers, Async Thunks.

Performance: Memoization, React.memo, useMemo, useCallback.

Code Splitting & Lazy Loading.

Mini Project: Todo App with Redux Toolkit and performance optimizations.

Phase 5: Testing & Deployment

Testing: Jest, React Testing Library, writing unit tests.

Deployment: Build process, deploying on Netlify/Vercel.

Mini Project: Write tests for components and deploy a sample app.

Optional: TypeScript & Advanced Patterns

TypeScript with React: Typing props, state, hooks.

Advanced Patterns: HOC, Render Props, Compound Components.

Mini Project: Build a reusable component library with TypeScript.

Diagrams and Flowcharts: Virtual DOM, Component Lifecycle, Redux Flow will be included.